

Mortgage Loan Management System (MLMS)

Cloud-Native Microservices Case Study (Azure Integrated)

1. Cloud Platform Overview

All services are deployed and managed on **Microsoft Azure**, using **PaaS-first architecture** to reduce infrastructure complexity for freshers.

2. Azure Services Mapping (What & Why)

Azure Service	Used For	Why It Is Used
Azure App Service	Hosting APIs & Angular UI	No server management, easy CI/CD
Azure SQL Database	Mortgage business data	Relational, ACID compliance
Azure Service Bus	Service-to-service communication	Reliable async messaging
Azure Blob Storage	Document storage	Cost-effective object storage
Azure Key Vault	Secrets management	Secure credentials
Azure Cosmos DB	Identity users	High availability & scalability

3. Updated Microservices Architecture (Azure

High-Level Architecture

Angular (App Service - Web)

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| JWT

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API Gateway (Optional - App Service)

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Identity Service	Cosmos DB
Customer Service	Azure SQL
Loan Service	Azure SQL

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Azure Service Bus (Topics / Queues)

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Blob Storage / Key Vault

4. Azure App Service – Deployment Strategy

What is Deployed

Component	Deployment
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Angular UI	App Service (Static Web / Web App)
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Identity API	App Service
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Customer API	App Service
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Loan API	App Service
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Key Learning for Students

- App Service plans
- Environment variables
- App Settings
- Connection strings
- Deployment slots (conceptual)

5. Azure SQL Database – Business Data

Used By

- **Customer Service**
- **Loan Service**

EF Core Code-First Flow

Entity → DbContext → Migration → Azure SQL

Tables

- Customers
- LoanApplications

Why Azure SQL?

- Familiar SQL Server engine
 - Managed backups
 - Scales easily
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6. Azure Cosmos DB – Identity Users

Used By

- **Identity Service**

Why Cosmos DB for Identity?

- High availability
- Low latency
- Schema-flexible (JSON)
- Ideal for authentication workloads

Stored Data

```
{  
  "id": "user-guid",  
  "username": "john.doe",  
  "passwordHash": "...",  
  "role": "Customer"
```

}

7. Azure Service Bus – Service to Service Communication

Communication Pattern

Asynchronous, Event-Driven

Example Flow

1. Loan Service → Loan Submitted
2. Event published to Service Bus
3. Customer Service consumes event
4. Notification / Audit triggered

Simple Events

- LoanSubmittedEvent
- LoanApprovedEvent
- LoanRejectedEvent

Why Service Bus?

- Guaranteed delivery
 - Decoupled services
 - Retry & dead-letter support
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8. Azure Blob Storage – Document Handling

Used For

- Property documents (future scope)
- ID proof uploads (mock/demo)

What Students Learn

- Container creation
- Upload blobs

- Secure access using SAS tokens
 - Private containers
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9. Azure Key Vault – Security & Secrets

Secrets Stored

- SQL connection strings
- Cosmos DB keys
- Service Bus connection strings
- JWT signing key

Access Pattern

App Service → Managed Identity → Key Vault

Key Learning

- Never store secrets in appsettings.json
 - Centralized secret management
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10. JWT Security (Azure-Ready)

Authentication Flow

1. User logs in (Angular)
2. Identity Service validates user (Cosmos DB)
3. JWT generated
4. Token stored in browser
5. Angular interceptor sends token
6. APIs validate token

Role-Based Authorization

[Authorize(Roles = "LoanOfficer")]

11. Updated User Stories (Azure Included)

Customer

- Apply for mortgage loan
- Upload property document (Blob)
- View loan status

Loan Officer

- View applications
- Approve / reject loans
- Trigger Service Bus events

Admin

- Manage users
 - Configure roles
 - Monitor services
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12. 5-Day (40 Hours) Azure-Integrated Plan

Day 1 – Architecture & Azure Basics

- Azure overview
- App Service creation
- SQL & Cosmos DB setup

Day 2 – Identity Service

- Cosmos DB integration
- JWT + Roles
- Key Vault secrets

Day 3 – Customer & Loan Services

- EF Core with Azure SQL
- Service Bus integration

Day 4 – Angular + Azure

- Angular deployment
- JWT interceptor
- Secure API calls

Day 5 – Integration & Review

- End-to-end testing
- Role validation
- Azure monitoring overview

13. Expected Enterprise Readiness Outcome

After completing this case study, students will:

- Deploy **real microservices on Azure**
- Use **Azure SQL, Cosmos DB, Service Bus**
- Secure applications using **Key Vault**
- Understand **cloud-native architecture**